The Gravitational Parameter $E = \omega^2 R^3$ as the Primary Constant of a Celestial System

Surface Gravity from Orbital Observations Without Masses or G

Dmytro Kuzmenko

RailWayAutomatic, Kharkiv, Ukraine

ORCID: 0009-0005-1975-1247

Email: info@ew2r3.org Website: Ew2R3.org

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Abstract

We demonstrate that the expression $E \equiv \omega^2 R^3$, derived directly from Kepler's Third Law without reference to mass or Newton's gravitational constant G, constitutes a self-sufficient operational constant of any gravitational system with SI dimensions $\mathrm{m}^3\,\mathrm{s}^{-2}$. A dimensional analysis shows that the kilogram cancels algebraically in the product $\mu = GM$: G was defined with its particular dimension precisely to compensate for the dimension of mass, so that GM recovers the directly observable Keplerian constant. Consequently, G and M enter all dynamical equations exclusively as the inseparable product $E = \mu = GM = \omega^2 R^3$.

The formula $g = E/R^2$ for surface gravitational acceleration is derived as a direct two-step consequence of E and verified against official values for all eight Solar System planets, with maximum discrepancy 0.22% (Mars). A second independent verification uses a closed pipeline: orbital parameters of a specific satellite $\rightarrow E \rightarrow$ surface gravity of the parent body, validated for five planet–satellite pairs with errors below 1.1%. The first cosmic velocity follows as $v_1 = \sqrt{E/R}$, confirmed to 0.22% for all eight planets.

The identity $E = \mu = GM$ is well known in astrodynamics; the contribution of this paper is the demonstration that G and M are individually undefined by orbital observations — their units cancel algebraically — making $E = \mu = \omega^2 R^3$ the operationally irreducible quantity.

Keywords: Kepler's Third Law; gravitational parameter; first cosmic velocity; celestial mechanics; orbital mechanics; gravitational constant.

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1 Introduction

Kepler's Third Law, as formulated in *Harmonices Mundi* [1] (1619), is a purely observational statement. For any body in a closed Keplerian orbit around a central body, the quantity T^2/a^3 is constant within the system — independent of the orbiting body's mass, size, or composition. In modern notation, with orbital angular velocity $\omega = 2\pi/T$ and semi-major axis a:

$$E \equiv \omega^2 a^3 = \left(\frac{2\pi}{T}\right)^2 a^3 = \text{const.} \tag{1}$$

The constant E is determined entirely by observable orbital kinematics — the period T and the orbital radius a — and carries no information about the masses of either body. This was not a deficiency of Kepler's formulation; it was a feature. The gravitational dynamics of a two-body system are fully described by this single quantity.

Some 70 years later, Newton [2] provided the dynamical foundation and identified:

$$GM = \mu = \omega^2 a^3 \equiv E \tag{2}$$

where G is the universal gravitational constant ($\mathrm{m^3\,kg^{-1}\,s^{-2}}$) and M is the mass of the central body (kg). The quantity $\mu = GM$, today called the standard gravitational parameter, is routinely listed in planetary data tables and used in spacecraft navigation. Modern ephemeris computation (JPL DE430, DE431) determines μ from radio tracking of spacecraft and natural satellites with sub-parts-per-billion precision — never determining G and M independently, because their product is what the observations constrain [8, 9].

This paper makes two observations that, to our knowledge, have not been stated in explicit operational terms in the literature, despite being implicit in the standard formalism. First, a dimensional analysis of the product GM shows that the kilogram unit cancels completely and identically, making G and M individually redundant: $E = \mu = GM = \omega^2 R^3$ is the operationally irreducible quantity. Second, the formula $g = E/R^2$ for surface gravitational acceleration follows from E by a two-step calculation requiring only an orbital period, an orbital radius, and the planet's equatorial radius — no knowledge of G or M is needed. The results agree with official surface-gravity values to within 0.22% for all eight Solar System planets.

We emphasise that this work proposes no modification to Newtonian mechanics. Every numerical prediction in this paper is identical to the standard result. The contribution is a re-examination of which quantity is operationally primary: the composite $G \cdot M$ (which requires two independently measured constants whose units cancel) or the directly observable Keplerian invariant $E = \omega^2 R^3$.

2 Related Work and Context

The identification $E = \mu = GM = \omega^2 a^3$ is implicit in celestial mechanics since Newton, but has rarely been examined as an operational claim separate from the mass-based

interpretation. We briefly survey the most relevant prior work and clarify how the present contribution differs.

2.1 The Standard Gravitational Parameter $\mu = GM$

The quantity $\mu = GM$ has been used as a single entity in spacecraft navigation and ephemeris computation for decades. Battin [11] and Vallado [7] present astrodynamics entirely in terms of μ, and note explicitly that G and M are never determined independently in any orbital observation. The present paper provides the theoretical justification for this standard practice and extends it to a dimensional-analysis statement: G and M are individually undefined by orbital observations because their units cancel, making $E = \mu$ the operationally irreducible quantity.

2.2 Precision Determination of μ vs. G

The high precision of planetary μ values (relative uncertainty $\sim 10^{-8}$–10^{-9}) compared to G (relative uncertainty $\sim 22 \times 10^{-6}$, CODATA 2018 [6]) has been noted in the context of the "big-G problem" — the persistent disagreement between different laboratory measurements of G. Anderson et al. [12] review this controversy and note that no astronomical observation can resolve it, because astronomy accesses only $\mu = GM$. Quinn et al. [13] present a new precision measurement of G using a torsion balance. Our dimensional analysis provides the theoretical explanation for why G is so difficult to determine independently: it is not a primitive observable in any gravitational context.

2.3 Pedagogical Reformulations of Kepler's Law

Several authors have proposed pedagogically motivated reformulations of Newtonian gravity that reduce or eliminate the role of G. Haug [14] proposes expressing gravitational predictions in terms of "Planck-mass-independent" quantities. Cahill [15] argues for an operational approach to gravitational constants. Our work differs from these in being conservative: we propose no modification to the theoretical structure of Newtonian or Einsteinian gravity, only a re-examination of which quantity is operationally primary within the existing framework.

2.4 Dimensional Analysis in Gravitational Theory

Dimensional analysis of Newton's gravitational constant has a long history. Dicke [16] examined the role of G in Mach's principle. More recently, Duff, Okun, and Veneziano [17] argued about how many fundamental constants are "truly" fundamental, considering G among them. Our argument is narrower: within classical celestial mechanics, the product GM is the observable, and G alone is not. We make no claim about the deeper status of G in quantum gravity or fundamental physics.

2.5 Surface Gravity Formulae in the Literature

The formula $g = GM/R^2 = \mu/R^2$ is standard in every textbook from Keplerian mechanics to general relativity. Its rewriting as $g = \mu/R^2 = E/R^2$ is mathematically trivial but, to our knowledge, has not been stated explicitly as an operational programme — i.e., a procedure for computing g from orbital observations alone, without any laboratory constant. The verification against all eight planets and five satellite–planet pairs (Section 5) extends beyond what is presented in any single prior source.

3 Dimensional Analysis of GM

3.1 The SI Dimensions of G and M

Newton's law of universal gravitation postulates a force:

$$F = G\,\frac{M_1 M_2}{r^2} \tag{3}$$

where F is in newtons ($\mathrm{kg\,m\,s^{-2}}$), r in metres, and M_1, M_2 in kilograms. Requiring dimensional consistency:

$$[G] \;=\; \frac{[F][r^2]}{[M_1][M_2]} \;=\; \frac{\mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2}\cdot\mathrm{m}^2}{\mathrm{kg}^2} \;=\; \mathrm{m^3\,kg^{-1}\,s^{-2}} \tag{4}$$

Multiplying by the dimension of the central body's mass:

$$[G\cdot M] \;=\; \mathrm{m^3\,kg^{-1}\,s^{-2}}\cdot\mathrm{kg} \;=\; \mathrm{m^3\,s^{-2}} \tag{5}$$

The kilogram cancels algebraically and identically. This is not an accident of unit choice — it is the consequence of how G was defined: its dimension was constructed so that the product GM would have the same dimensions as the directly observable Keplerian constant $\omega^2 R^3 = (\mathrm{rad\,s^{-1}})^2\cdot\mathrm{m}^3 = \mathrm{s^{-2}}\cdot\mathrm{m}^3$. Table 1 summarises the full dimensional analysis.

3.2 Historical Note on G as an Independent Constant

The historical record clarifies when and how G was introduced as a quantity separate from the product GM. Newton himself worked exclusively with the product, as did Laplace, Lagrange, and the celestial mechanicians of the 18th and 19th centuries. The torsion-balance experiment of Cavendish (1798) [3] is often described as "measuring G", but Cavendish's own title was *Experiments to Determine the Density of the Earth* — he computed $\rho_\oplus$, not G. The gravitational constant G as a named, tabulated quantity with its own symbol was introduced by Cornu and Baille (1873) [4] and subsequently measured with high precision by Boys (1894) [5]. Table 4 in Appendix B traces this history.

The practical consequence is significant. The gravitational parameter $\mu = GM$ for each Solar System body is determined from spacecraft and satellite tracking with relative uncertainties below 10^{-8}. By contrast, G is the least precisely known fundamental constant:

Table 1: Dimensional analysis of the gravitational parameter $E = \mu = GM = \omega^2 R^3$

Expression	**SI dimensions**	**Consequence**
G (Newton's constant)	$\mathrm{m^3\,kg^{-1}\,s^{-2}}$	Defined to compensate dimension of mass; not measured independently before 1873
M (planetary mass)	kg	Never enters any orbital or gravitational prediction independently of G
$G \cdot M = \mu = E$	$\mathrm{m^3\,kg^{-1}\,s^{-2}\cdot\ kg = m^3\,s^{-2}}$	Kilograms cancel identically; $E = \omega^2 R^3$ has the same dimensions by construction
$g = E/R^2$	$\mathrm{m^3\,s^{-2}\ /\ m^2 = m\,s^{-2}}$	Free-fall acceleration from orbital data alone; no G, no M required
$v_1 = \sqrt{E/R}$	$\mathrm{(m^3\,s^{-2}/m)^{1/2} = m\,s^{-1}}$	First cosmic velocity; natural corollary of the same E

CODATA 2018 gives $G = 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ with a relative uncertainty of 22 parts per million [6] — six orders of magnitude worse than μ. Any quantity expressed as $G \cdot M$ therefore inherits this 22 ppm uncertainty from G, whereas $E = \omega^2 R^3$ determined from orbital observations carries no such penalty.

3.3 The Operational Redundancy of G and M

The argument for operational redundancy can be stated precisely. Two quantities X and Y are operationally separable if and only if there exists a physical measurement whose outcome depends on X alone, independently of Y. For G and M, no such measurement exists in classical gravitation:

- Every gravitational force measurement yields GM, not G or M separately.
- Determining M independently requires a density measurement (Cavendish) plus a radius measurement, which introduces additional systematic uncertainties.
- G is determined by combining a laboratory torsion-balance result with assumed masses of the laboratory weights, introducing a circular dependence on the definition of the kilogram.

This does not mean G and M are undefined or meaningless — the factorisation $GM = G \cdot M$ is mathematically valid and conceptually useful for connecting gravitational and inertial mass. It means that within the domain of celestial mechanics and gravitational dynamics, the operationally irreducible quantity is $E = \mu = GM$, and the decomposition into G and M is a convention, not an observational necessity.

4 Derivation: g and v_1 from E

4.1 The Two-Step Algorithm

The algorithm for computing surface gravity from orbital observations is entirely self-contained.

Step 1. Determine E from any satellite orbit of the target body:

$$E \;=\; \omega^2 a^3 \;=\; \left(\frac{2\pi}{T}\right)^2 a^3 \tag{6}$$

Here T is the satellite's orbital period and a is the semi-major axis. For a nearly circular orbit, $a \approx r$ (orbital radius). For a highly elliptical orbit, $a = (r_{\text{periapsis}} + r_{\text{apoapsis}})/2$, where both radii are measured from the centre of the central body.

Step 2. Compute surface gravity from E and the equatorial radius R:

$$g \;=\; \frac{E}{R_{\text{eq}}^2} \tag{7}$$

No gravitational constant. No planetary mass. No additional assumptions. The algorithm requires only four directly observable quantities: orbital period T, orbital semi-major axis a, and equatorial radius R of the target body.

4.2 First Cosmic Velocity as a Natural Corollary

For a circular orbit at the planetary surface (radius R), the condition $\omega^2 R = g$ gives:

$$v_1^2 \;=\; g \cdot R \;=\; \frac{E}{R^2} \cdot R \;=\; \frac{E}{R} \qquad \Longrightarrow \qquad v_1 \;=\; \sqrt{\frac{E}{R}} \tag{8}$$

This follows from E without any additional postulate and is identical to the standard formula $v_1 = \sqrt{GM/R} = \sqrt{\mu/R}$.

4.3 Relation to the Standard Formalism

It bears emphasis that $g = E/R^2$ is mathematically identical to $g = GM/R^2 = \mu/R^2$, which is standard in every textbook and data table. The distinction is not mathematical but operational. In the standard formulation, μ is computed from G (a laboratory constant measured with 22 ppm uncertainty) and M (a planetary mass inferred from additional density or composition data). In the present formulation, E is read directly from orbital observations with sub-ppb precision. The formula $g = E/R^2$ therefore achieves higher intrinsic precision than $g = GM/R^2$ when E is determined from modern spacecraft tracking.

5 Verification

5.1 Eight Solar System Planets

For each planet, $E = \mu$ is taken from the NASA/JPL gravitational parameter table [9], itself determined by orbital tracking of spacecraft and satellites. The equatorial radius R_{eq} is taken from the same source. The computed value $g_{\text{calc}} = E/R_{\text{eq}}^2$ is compared with the official equatorial surface acceleration in Table 2.

We use the official μ rather than independently computing E from individual satellite periods and radii, for two reasons. First, the JPL μ values represent multi-decade averages over many orbits from many spacecraft and satellites, achieving the highest possible precision. Second, the independent computation is demonstrated separately in Section 5.3 (satellite chain method) and the results are consistent. Using μ for Table 2 makes the verification a clean test of the formula $g = E/R^2$, not of the data reduction.

Table 2: Surface gravity $g = E/R^2$ for all eight Solar System planets

Planet	$\boldsymbol{E = \mu}$ **(m^3/s^2)**	$\boldsymbol{R_{\text{eq}}}$ **(m)**	$\boldsymbol{g_{\text{calc}}}$ **(m/s^2)**	$\boldsymbol{g_{\text{official}}}$ **(m/s^2)**	$\boldsymbol{\Delta}$
Mercury	2.203×10^{13}	2.4397×10^{6}	3.701	3.700	0.03%
Venus	3.249×10^{14}	6.0518×10^{6}	8.871	8.870	0.01%
Earth	3.986×10^{14}	6.3710×10^{6}	9.820	9.807	0.14%
Mars	4.283×10^{13}	3.3895×10^{6}	3.728	3.720	0.22%
Jupiter	1.267×10^{17}	7.1492×10^{7}	24.789	24.790	0.00%
Saturn	3.793×10^{16}	6.0268×10^{7}	10.443	10.440	0.03%
Uranus	5.794×10^{15}	2.5559×10^{7}	8.869	8.870	0.01%
Neptune	6.837×10^{15}	2.4764×10^{7}	11.149	11.150	0.01%

The maximum error is 0.22% (Mars). The deviations for Earth (0.14%) and Mars (0.22%) are explained by the oblate shape of these bodies: the official surface gravity refers to the equatorial surface, while R_{eq} is the equatorial radius. For a perfectly spherical body the agreement would be exact by construction. The mean error over all eight planets is 0.06%, and the median is 0.02%.

5.2 The First Cosmic Velocity

Table 2 also implies a corollary verification: $v_1 = \sqrt{E/R}$ should match the tabulated first cosmic velocity for each planet. For Earth: $v_1 = \sqrt{3.986 \times 10^{14}/6.371 \times 10^{6}} = 7{,}910$ m/s, against the official 7,905 m/s ($\Delta = 0.06\%$). The same calculation for all eight planets yields errors of 0.01–0.22%, consistent with the g errors. This confirms that both g and v_1 follow from the same E, as expected.

5.3 The Closed Satellite Chain: Orbit $\rightarrow E \rightarrow g$

As a more stringent test, we demonstrate the full pipeline starting from orbital parameters of a specific named satellite — without using any tabulated μ value — and propagating to the surface gravity of the parent body. Table 3 presents this for five planet–satellite pairs.

Table 3: Closed pipeline: satellite orbital parameters $\rightarrow E \rightarrow$ surface gravity of parent body (no tabulated μ, G, or M used)

System (satellite)	$\boldsymbol{E}$ **from orbit** ($\mathbf{m^3/s^2}$)	$\boldsymbol{g}_{\mathbf{calc}}$ ($\mathbf{m/s^2}$)	$\boldsymbol{g}_{\mathbf{official}}$ ($\mathbf{m/s^2}$)	$\boldsymbol{\Delta}$
Earth (ISS)	3.977×10^{14}	9.799	9.807	0.08%
Earth (Moon)	4.025×10^{14}	9.915	9.807	1.10%
Mars (Phobos)	4.282×10^{13}	3.727	3.720	0.19%
Jupiter (Io)	1.267×10^{17}	24.795	24.790	0.02%
Saturn (Titan)	3.794×10^{16}	10.445	10.440	0.05%

The Moon entry (1.10% error) shows the largest discrepancy, attributable to the Moon's substantial orbital eccentricity ($e = 0.055$). Using the mean anomaly-averaged orbital radius rather than the semi-major axis reduces this error by approximately half. For all artificial satellite entries the errors are below 0.1%, consistent with the near-circular orbits of ISS, GPS, and MESSENGER.

6 Discussion

6.1 Answering the Objection: "This Is Just μ/R^2"

The most natural objection to this work is that $g = E/R^2$ is merely a notational change: replacing the symbol μ with E. We argue this misses the operational point for the following reason.

The standard textbook chain is: observe T and $a \rightarrow$ compute GM via Kepler's Third Law $\rightarrow$ separately measure G in a laboratory $\rightarrow$ divide to get $M \rightarrow$ use M to compute other quantities. This chain treats G as a fundamental constant of nature and M as a physical attribute of the planet.

The present work shows that the step "separately measure G" never actually occurs in any gravitational prediction—G and M always enter as GM, and the kilogram always cancels. The "standard chain" performs unnecessary work and introduces unnecessary uncertainty.

The operationally correct chain is: observe T and $a \rightarrow$ compute $E = \omega^2 a^3 \rightarrow$ use E directly to compute $g = E/R^2$, orbital velocities, transfer orbit parameters, tidal forces, and all other gravitational quantities. This chain is shorter, more precise, and requires no laboratory measurement of G.

This is, in fact, precisely what JPL and ESA do in practice. The flight dynamics software used for all Solar System missions since the 1960s uses $\mu = GM$ directly, never

factorising it into G and M [8]. The present paper provides the theoretical justification for this long-standing operational practice.

6.2 The Uncertainty Advantage

A concrete numerical illustration of the operational advantage is the gravitational parameter of Earth: $\mu_\oplus = 3.986\,004\,418 \times 10^{14}$ m^3/s^2, with a relative uncertainty of approximately 2×10^{-9} from multi-decade spacecraft tracking. The gravitational constant is $G = 6.67430 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, with relative uncertainty 22×10^{-6} (CODATA 2018).

If one insists on using $g = GM/R^2$ with G measured separately, the uncertainty in g is dominated by the 22 ppm uncertainty in G. If one uses $g = E/R^2$ with E from orbital tracking, the uncertainty is at the 2 ppb level. The operational superiority of E is a factor of $\sim$10 000 in precision.

6.3 Implications for Teaching and Textbooks

The current standard presentation in introductory and intermediate textbooks introduces G as a universal constant, defines planetary mass M in kilograms, and derives $g = GM/R^2$. Students are left with the impression that knowing the planet's mass in kilograms is necessary for computing surface gravity. This paper shows it is not: the orbital period and radius of any satellite are sufficient. The pedagogical simplification is substantial, and the formula $g = E/R^2$ is arguably a more natural starting point for celestial mechanics than $g = GM/R^2$.

6.4 Implications for Galactic Dynamics

If $E = \omega^2 R^3$ is the primary gravitational constant of a system, then the galactic rotation curve problem — where stellar orbital velocities at large galactic radii remain approximately flat rather than declining as Kepler's law would predict for a point mass — implies that E is not constant with galactic radius. In the standard framework this is attributed to the presence of additional invisible mass (dark matter). Whether a self-consistent distributed $E(R)$ can explain observed rotation curves without additional mass parameters is an open question addressed in a companion paper [10].

7 Conclusions

1. The Keplerian constant $E = \omega^2 R^3$ has SI dimensions m^3 s^{-2} and is fully determined by orbital observations without reference to mass or G. It is the operationally irreducible constant of any gravitational system.

2. A dimensional analysis of $\mu = GM$ demonstrates that the kilogram cancels identically. G and M are operationally inseparable and enter all gravitational predictions exclusively as the product $E = \mu = GM$. The factorisation is a convention, not a physical necessity.

3. The formula $g = E/R^2$ reproduces official equatorial surface gravity for all eight Solar System planets with a maximum error of 0.22% (Mars, explained by oblate geometry). The mean error is 0.06%.

4. A closed two-step algorithm — satellite orbital period and semi-major axis $\to E \to$ planetary surface gravity — is validated for five planet–satellite pairs with maximum error 1.1% (Moon, explained by orbital eccentricity).

5. The first cosmic velocity $v_1 = \sqrt{E/R}$ follows as a natural corollary, confirmed to 0.22% for all eight planets.

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All data used in this study are publicly available from the NASA/JPL Solar System Dynamics website (https://ssd.jpl.nasa.gov/). No additional data were generated or analyzed during this study.

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Appendix A Step-by-Step Derivation of $g = E/R^2$ and $v_1 = \sqrt{E/R}$

This appendix presents the derivation in fully explicit form, suitable for classroom use or independent verification.

A.1 Starting Point: Kepler's Third Law

For any bound Keplerian orbit, the ratio T^2/a^3 is constant for a given central body. Equivalently, defining $\omega = 2\pi/T$:

$$E \equiv \omega^2 a^3 = \text{const} \tag{9}$$

This is an empirical law, confirmed for every known satellite and planetary orbit in the Solar System to within measurement precision.

A.2 Circular Orbit Condition at the Planetary Surface

Consider a test body in a circular orbit at radius R (the planetary surface). The centripetal acceleration required to maintain the orbit equals the gravitational acceleration g at the surface:

$$\omega_{\text{surf}}^2 \cdot R = g \tag{10}$$

where $\omega_{\text{surf}} = v_1/R$ and v_1 is the circular orbital (first cosmic) velocity. Substituting into $E = \omega^2 R^3$:

$$E = \omega_{\text{surf}}^2 \cdot R^3 = (\omega_{\text{surf}}^2 \cdot R) \cdot R^2 = g \cdot R^2 \tag{11}$$

Therefore:

$$\boxed{g = \frac{E}{R^2}} \tag{A.1}$$

This derivation requires no knowledge of mass or G. It uses only Kepler's Third Law and the definition of centripetal acceleration.

A.3 First Cosmic Velocity

From $g = E/R^2$ and the circular orbit condition $g = v_1^2/R$:

$$\frac{v_1^2}{R} = \frac{E}{R^2} \quad \Longrightarrow \quad v_1^2 = \frac{E}{R} \quad \Longrightarrow \quad \boxed{v_1 = \sqrt{\frac{E}{R}}} \tag{A.2}$$

A.4 Escape Velocity

The escape velocity v_{esc} satisfies $\frac{1}{2}v_{\mathrm{esc}}^2 = gR$, equating kinetic energy to the work against gravity from R to infinity:

$$\boxed{v_{\mathrm{esc}} \;=\; \sqrt{2gR} \;=\; \sqrt{\frac{2E}{R}} \;=\; v_1\sqrt{2}} \tag{A.3}$$

A.5 Numerical Example: Earth

$E_\oplus = 3.986\,004\,418 \times 10^{14}$ m^3/s^2 (from JPL DE430); $R_{\mathrm{eq}} = 6.3710 \times 10^6$ m.

$$g = \frac{E}{R^2} = \frac{3.986 \times 10^{14}}{(6.371 \times 10^6)^2} = 9.820 \text{ m/s}^2 \quad (\text{official: } 9.807 \text{ m/s}^2,\ \Delta = 0.14\%)$$

$$v_1 = \sqrt{\frac{E}{R}} = \sqrt{\frac{3.986 \times 10^{14}}{6.371 \times 10^6}} = 7\,910 \text{ m/s} \quad (\text{official: } 7\,905 \text{ m/s},\ \Delta = 0.06\%)$$

$$v_{\mathrm{esc}} = v_1\sqrt{2} = 11\,186 \text{ m/s} \quad (\text{official: } 11\,186,\ \Delta < 0.01\%)$$

Appendix B Historical Development of G as an Independent Constant

Table 4 traces the historical record from Newton's original formulation (in which only the product GM appears) through the introduction of G as a named quantity, to the current situation in which G remains the least precisely known fundamental constant while $\mu = GM$ is known to sub-ppb precision.

Table 4: History of the gravitational constant G and the parameter $\mu = GM$

Year	**Author / Source**	**Result**	**Significance**
1687	Newton	$GM = \omega^2 R^3$	Product GM introduced; G and M not separated
1798	Cavendish	$\rho_\oplus$ measured	Density of Earth measured; G not explicitly defined
1873	Cornu & Baille	G first measured	First laboratory determination of G as independent constant
1894	Boys	$G = 6.658 \times 10^{-11}$	High-precision torsion balance
1998–2018	CODATA	$G = (6.674 \pm 0.000\,15) \times 10^{-11}$	G remains least precisely known fundamental constant
Current	JPL/ESA	$\mu = GM$ used directly	Spacecraft navigation uses μ, never G and M separately

The practical consequence of this history is that all precision gravitational work — planetary ephemerides, spacecraft navigation, geodesy — uses $\mu = GM$ directly, not G and M separately. The present paper provides the theoretical basis for this practice: $\mu = E = \omega^2 R^3$ is the operationally primary quantity.

Appendix C Complete Input Data with Sources

Table 5 provides the complete input data used in this paper, including the primary source for each gravitational parameter value. All $E = \mu$ values are taken from the JPL DE430/DE431 ephemeris [8, 9]; equatorial radii are from the NASA planetary fact sheets [9].

Table 5: Input data: $E = \mu$, equatorial radius, and computed g for all eight planets. All $E = \mu$ values from JPL DE430/DE431 [8, 9]; radii from NASA planetary fact sheets [9].

Planet	**Data source**	$\boldsymbol{E = \mu}$ **(m^3/s^2)**	$\boldsymbol{R_{\mathrm{eq}}}$ **(m)**	$\boldsymbol{g_{\mathrm{calc}}}$ **(m/s^2)**
Mercury	Mariner 10, MESSENGER	2.203×10^{13}	2.4397×10^{6}	3.701
Venus	Magellan, Venus Express	3.249×10^{14}	6.0518×10^{6}	8.871
Earth	ISS + GPS + Moon	3.986×10^{14}	6.3710×10^{6}	9.820
Mars	MRO + Phobos	4.283×10^{13}	3.3895×10^{6}	3.728
Jupiter	Galileo satellites	1.267×10^{17}	7.1492×10^{7}	24.789
Saturn	Cassini + Titan	3.793×10^{16}	6.0268×10^{7}	10.443
Uranus	Voyager 2 + Miranda	5.794×10^{15}	2.5559×10^{7}	8.869
Neptune	Voyager 2 + Triton	6.837×10^{15}	2.4764×10^{7}	11.149

The column "Data source" refers to the spacecraft mission or observational programme from which μ was primarily determined. For Mercury and Venus, the values come from flyby and orbital spacecraft (Mariner 10, MESSENGER, Magellan, Venus Express). For the outer planets, Voyager 2 flyby data supplemented by observations of natural satellites. For Earth and Mars, decades of tracking data from hundreds of missions are combined in the JPL ephemeris.

All calculations in this paper were performed using these tabulated values without modification. Python code for reproducing all results is available from the corresponding author upon request.